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Comparison of E-Waste Management Rules and Regulations

Table 1: Comparison of E-Waste Management Rules and Regulations

India • USA • Australia • Russia • Japan

Country	Main Rules / Laws	Key Responsibilities	Disposal / Recycling Approach
India	E-Waste (Management) Rules, 2022; amendments apply.	Producers have Extended Producer Responsibility (EPR); manufacturers, producers, refurbishers, dismantlers and recyclers are regulated.	E-waste is handled through registered channels and recycling targets/EPR mechanisms.
USA	No single comprehensive federal e-waste law; RCRA and other federal rules apply, while many states have their own electronics laws.	EPA regulates relevant hazardous wastes; states can impose additional electronics recycling requirements.	Recycling and disposal rules vary by state; 25 states plus D.C. currently have electronics recycling laws.
Australia	Recycling and Waste Reduction Act 2020; NTCRS and Product Stewardship Rules 2021.	Importers/manufacturers above thresholds have obligations; approved co-regulatory arrangements organize collection and recycling.	Households and small businesses can access free NTCRS collection for covered TVs and computers.
Russia	Federal waste legislation, Extended Producer Responsibility (EPR/РОП), and waste-classification rules.	Producers/importers have recycling obligations; reporting and utilization mechanisms are being strengthened.	Electronic equipment waste is classified and managed under waste rules, with growing emphasis on producer-funded utilization.
Japan	Home Appliance Recycling Law; Small Home Appliance Recycling Law.	Consumers pay applicable fees; retailers collect covered appliances; manufacturers/importers recycle them.	Strong take-back and recycling system for TVs, air conditioners, refrigerators/freezers, and washing machines/dryers; small appliances are also promoted for metal recovery.

Key Differences

India: Uses a national EPR-based framework with formal obligations for producers and recyclers.

USA: Has a more decentralized system, with major differences between federal and state requirements.

Australia: Uses a national product-stewardship scheme for televisions and computers with industry funding.

Russia: Uses broader waste legislation and EPR mechanisms; electronic waste is classified within the national waste system.

Japan: Has a clearly defined take-back model for major household appliances, with consumer fees and manufacturer recycling duties.

Conclusion

All five countries aim to reduce pollution and recover valuable materials from e-waste, but their systems differ in structure. India and Australia use strong producer-responsibility approaches, Japan combines consumer fees with retailer and manufacturer duties, and the USA relies heavily on state-level regulation. Russia is developing and strengthening producer responsibility and utilization mechanisms within its broader waste-management framework.

Sources: Government of India (MoEFCC); U.S. EPA; Australian DCCEEW; Government of Russia/Rosprirodnadzor; Japan Ministry of the Environment.